Reference-Based Alignment of Large Sequence Databases

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Motivation

- Defined over an alphabet Σ.
- Text documents.
- Biological sequences (DNA).
- Near homology search:
  - Deviation from Q does not exceed a threshold δ (δ ≤ 15%).

Q: TCTAGGGCA

…ACTTAGCTGTAGTCGTTCTATGGCATATGCATGCTGATCTCGTGCGTCATG…

String Matching

- Given:
  - S: collection of sequences defined over an alphabet Σ.
  - Q: query sequence defined over Σ.
  - D: similarity measure.
- Find the most similar subsequence in S.

Our focus: DNA

- S: a set of DNA sequences.
- Q: DNA sequence
  - with a small deviation from the database match.
    - within δ $|Q|$, for $\delta \leq 15\%$.
  - can be large (up to 10,000 nucleotides).

The Edit Distance

- Initialization:

		A	C	T	G
	0	1	2	3	4
A	1				
T	2				
C	3				

The Edit Distance

- First column:

- Match = 0
- In/del/sub = 1

		A	C	T	G
	0	1	2	3	4
A	1	0			
T	2	1			
C	3	2			

The Edit Distance

- Second column:

		A	C	T	G
	0	1	2	3	4
A	1	0	1		
T	2	1	1		
C	3	2	1		

The Edit Distance

- Final Matrix:

		A	C	T	G
	0	1	2	3	4
A	1	0	1	2	3
T	2	1	1	1	2
C	3	2	1	2	2

The Edit Distance

- Alignment Path:

A = A – T C

B = A C T G

		A	C	T	G
	0	1	2	3	4
A	1	0	1	2	3
T	2	1	1	1	2
C	3	2	2	2	2

The Edit Distance: Subsequence matching

- Initialization:

		A	C	T	G
	0	0	0	0	0
A	1				
T	2				
C	3				

The Edit Distance: Subsequence matching

- Final Matrix:

		A	C	T	G
	0	0	0	0	0
A	1	0	1	1	1
T	2	1	1	1	2
C	3	2	2	2	2

The Edit Distance: Subsequence matching

- One path:

A = A T C

B = A C T G

		A	C	T	G
	0	0	0	0	0
A	1	0	1	1	1
T	2	1	1	1	2
C	3	2	1	2	2

Smith-Waterman [Smith&Waterman et al. 1981]

- Is a similarity measure used for local alignment:
  - Match can be a subsequence of the query sequence.
- Define three penalties:
  - match, mismatch, gap.
  - Scoring parameters are defined by the user.
- Example:
  - *A* = ATC and *B* = TATTCG
  - match = 2, mismatch = -1, gap = -1.

Smith-Waterman

- Initialization:

		T	A	T	T	C	G
	0	0	0	0	0	0	0
A	0						
T	0						
C	0						
A	0						

Smith-Waterman

- First column:

		T	A	T	T	C	G
	0	0	0	0	0	0	0
A	0	-1					
T	0	2					
C	0	1					
A	0	0					

Smith-Waterman

- First column:

		T	A	T	T	C	G
	0	0	0	0	0	0	0
A	0	0					
T	0	2					
C	0	1					
A	0	0					

Smith-Waterman

- Second column:

		T	A	T	T	C	G
	0	0	0	0	0	0	0
A	0	0	2				
T	0	2	1				
C	0	1	1				
A	0	0	3				

Smith-Waterman

- Final Matrix:

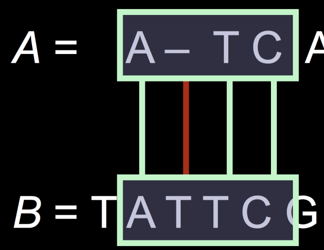
		T	A	T	T	C	G
	0	0	0	0	0	0	0
A	0	0	2	1	0	0	0
T	0	2	1	2	3	2	1
C	0	1	1	1	2	5	4
A	0	0	3	2	1	4	4

Smith-Waterman

- Detect highest value:

		T	A	T	T	C	G
	0	0	0	0	0	0	0
A	0	0	2	1	0	0	0
T	0	2	1	2	3	2	1
C	0	1	1	1	2	5	4
A	0	0	3	2	1	4	4

Smith-Waterman



- Alignment Path:

		T	A	T	T	C	G
	0	0	0	0	0	0	0
A	0	0	2	1	0	0	0
T	0	2	1	2	3	2	1
C	0	1	1	1	2	5	4
A	0	0	3	2	1	4	4